

# WESLEY JANES

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## EDUCATION & HONORS

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### Mississippi State University, Bagley College of Engineering

B.S. Aerospace Engineering, *Concentration in Astronautics* | GPA: 3.44 / 4.0

Shackouls Honors College | President's List / Dean's List | Eagle Scout

Starkville, MS

August 2021 – May 2026

**Relevant Coursework:** Aerothermodynamics, Aerospace Controls, Spacecraft Attitude Dynamics, Aerostructure Design, Orbital Mechanics, Fluid Mechanics, Spacecraft Design, Vibrations, Compressible Aerodynamics, Spacecraft Propulsion, Circuits, Computer Programming, Calculus, Physics.

## ENGINEERING EXPERIENCE

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### Applied Aerodynamics Research at Mississippi State University

*Undergraduate Research Assistant & Co-Author*

Starkville, MS

August 2024 – Present

- Engineered **16 advanced airfoil geometries** using **SolidWorks** and resin-based **additive manufacturing**, reducing prototype post-processing times by **24+ hours** and enabling rapid iteration of leading-edge modifications.
- Streamlined data collection** by conducting subsonic wind tunnel experiments and supporting analysis through **Monte Carlo** simulations with real-time **LabVIEW data acquisition**, ensuring high-fidelity aerodynamic profiling across multiple Reynolds numbers.
- Developed custom **MATLAB** post-processing scripts to automate lift, drag, and moment coefficients, cutting data reduction time by **75%**.
- Spearheaded **oil flow visualization studies** across multiple airfoil profiles to identify boundary surface flow behaviors, co-authoring a technical paper for **journal publication** (expected publication: **June 2026**).

### Xipiter Unmanned Aerial Systems Club

*CAD Team Lead*

Starkville, MS

January 2025 – Present

- Directed a multidisciplinary team of 14 student-engineers** in the structural design of a custom UAS airframe in **SolidWorks**, optimizing internal structural components to integrate **avionics systems**.
- Developed mission-critical **Python and C** scripts to automate flight **guidance, navigation, and control parameters**, directly enabling the aircraft to execute the designated competition mission profile.
- Supervised **composite airframe fabrication**, applying **GD&T** standards to guarantee precision integration between carbon fiber components and internal sheet metal mounts.

### Systems & Mission Design Analysis of the X-15 Hypersonic Vehicle

*Modeling & Simulation Lead*

Starkville, MS

August 2025 – December 2025

- Translated historical **mission requirements** into technical specifications to develop a **SolidWorks** airframe model with **98%** geometric accuracy.
- Performed **3D thermal-structural analysis** using ANSYS with an accuracy window of **±5%**, simulating temperature contours at Mach 8 to verify engineers of the 1960's **design requirements** and performance limits of the thermal ablative coating.
- Presented technical findings to **nationally recognized museums**, delivering a mission briefing that correlated modern computational results with historical flight data for public exhibition.

### Aerostructure Design of Stiffened Panels of Multiple Failure Modes

*Manufacturing Lead*

Starkville, MS

August 2025 – December 2025

- Engineered structural airframe models in **SolidWorks** and performed **Abaqus FEA** to identify critical stress concentrations, ensuring the assembly maintained a **1.5 factor of safety** under predicted failure loads.
- Optimized manufacturing process by applying **GD&T** principles to technical processes and overseeing **sheet metal fabrication process**, ensuring **cost-effective methods** for precision cutting, bending, drilling, and riveting.
- Achieved **95%** accuracy in results between UTM data, computational FEA, and theoretical calculations.

### Compressible Flow Analysis of Rocket Nozzles

*Software Analysis Lead*

Starkville, MS

August 2024 – December 2024

- Simulated converging/diverging nozzle performance in ANSYS, analyzing shockwave formations and choked flow to optimize contours at target altitudes; achieved a **90%** correlation between computational results and isentropic flow equations.
- Automated performance visualization via **MATLAB**, slashing data processing time by **80%** and enabling rapid iteration of mass flow calculations.

### The Lilly Company

*IT Assistant*

Memphis, TN

May 2023 – December 2023

- Optimized system deployment workflows by utilizing **AI/ML** and standardizing hardware configuration, resulting in a **50%** increase in technical setup efficiency for fleet-wide industrial mobile assets.
- Managed critical network infrastructure and data integrity, resolving **200+** service tickets to ensure seamless system functionality for regional logistical operations.

## TECHNICAL SKILLS

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**Software:** MATLAB, Python, C/C++, ANSYS, SolidWorks, LabVIEW, AutoCAD, Abaqus, STK, XFOIL, Git, Arduino, Linux, Microsoft Suite.

**Manufacturing:** Additive Manufacturing, CNC Machining, Composite Fabrication, Sheet Metal Fabrication, GD&T, Power Tools, Cost-Efficiency.